

Olist Order Fulfillment Analysis: Identifying Delivery Confirmation Gaps

Business Problem:

Analyzed 100K order to identify fulfillment breakdowns and delivery performance issues that could inform carrier operations decisions.

Key Findings:

Roughly 2.3% of all orders (2,341 of 99,441) fail at one or two distinct points in the pipeline, and both failure patterns are consistent across 2017 and 2018 rather than tied to a single event or season - pointing to systemic process gaps rather than one time issues.

- Post shipment confirmation gap: 1,107 orders (1.1%) show a confirmed carrier pickup date but no delivery confirmation date - the package left the warehouse but the delivery loop was closed.
- Post-approval fulfillment failures: 1,234 orders (1.2%) - 625 canceled and 609 marked unavailable - were approved (i.e. payment cleared) before failing indicating the breakdown happens after checkout, not at the point of sale.
- Order failure rates (shipped, confirmed, unavailable) were checked against customer state to rule out regional or carrier-specific causes. No state showed a disproportionate share of problem orders relative to its overall order volume, indicating these are platform-wide process issues rather than localized shipping/carrier problems.

Supporting Data:

Order Status Breakdown

Order Status	Count	% of total
Delivered	96,478	97.0%
Shipped	1,107	1.1%
Canceled	625	0.6%
Unavailable	609	0.6%
Invoiced	314	0.3%
Processing	301	0.3%
Created	5	<0.1%

Shipped	2	<0.1%
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Shipped but Unconfirmed Orders (1,107)

- 100% have a confirmed carrier pickup but no delivery confirmation
- Spread consistently across 2017 and 2018, not clustered in a single period, ruling out a one time event.
- Present in both 2017 and 2018, but not evenly: a sharp, isolated spike occurred in February-March 2018, accounting for a disproportionate share of that year's cases. No comparable spike occurred in 2017, suggesting a specific operational (i.e. system issue or carrier disruption) event rather than a recurring seasonal pattern.

Canceled Orders (625)

- Majority were approved before cancellation, indicating the issue occurs post checkout
- Data quality flag: a small number of "canceled" orders also carry a completed order customer delivery date, a status contradiction worth auditing in the source system.

Unavailable Orders (609)

- Majority were also approved prior to being marked unavailable, mirroring the canceled order pattern.
- Spread across 2017 and a substantial share of 2018 - not concentrated around a seasonal peak, suggesting an ongoing stock/inventory process issue rather than a holiday driven spike.

Regional Analysis

- To rule out a regional or carrier specific cause, problem orders (canceled, shipped, unavailable) were checked against customer state distribution. Sao Paulo (SP), the largest state by order volume at 42,0% of all orders, accounted for a proportional 40.4% of problem orders. This proportionality held consistent across other states as well, confirming that failures are platform wide rather than concentrated in a specific region or carrier network.

Recommendation:

- Audit delivery confirmation systems with carrier patterns; flag any orders with a carrier pickup date but no delivery confirmation 14+ days out for manual follow up.

- Review post approval fulfillment workflow since both cancellations are unavailability originate after payment is confirmed, not at checkout.
- Correct the canceled but delivered data contradiction at the source to preserve status field integrity for future reporting.
- Note: this analysis was cross-checked using two independent methods (Tableau and Google sheets pivot tables) to confirm consistency, an initial discrepancy was traced to a truncated data range in one pivot table and corrected before finalizing these figures.

Next Steps:

- Layer in order value data to estimate revenue impact of the 2,341 affected orders.
- Build a recurring dashboard view to monitor these two failure rates over time as new orders are processed.